

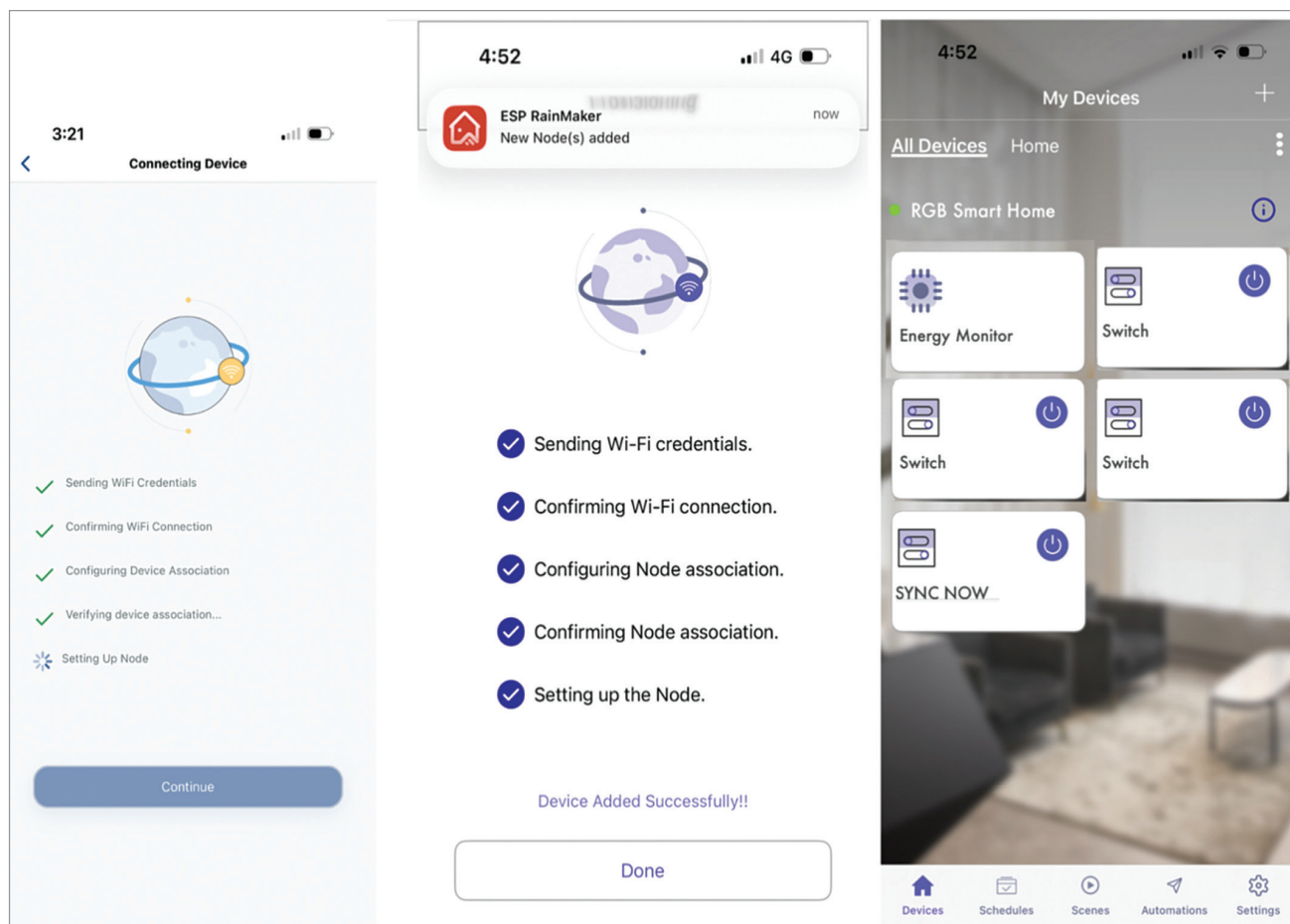


Fig. 9: RainMaker adding the device node and UI showing all sensors and devices for monitoring and control

```
void loop() {
  // read the voltage and then print via
  Serial.
  float voltage = voltageSensor.
  getRmsVoltage();
  Serial.print(voltage);
  Serial.print("V");
  Serial.println("-----");
  delay(1000);
}
```

Circuit and working

After calibration, complete the circuit as shown in Fig. 6. The system is built around the IndusBoard (U1), ACS712 current sensor (U2), ZMPT101B voltage sensor (U3), a 5V relay module (U4), and supporting components. A 5V DC power adaptor supplies the circuit.

The current sensor output connects to the IndusBoard via a voltage divider circuit. The

ZMPT101B is powered directly from the board, and its output pin connects to pin 5 or another pin as defined in the code. The load is connected between the current sensor's live output and the AC mains neutral.

The ACS712 operates at 5V logic, while the IndusBoard uses 3.3V logic. Therefore, a voltage-divider circuit with two resistors is required to reduce the signal level. During testing, resistor values of 10kΩ and 2kΩ were used. The code may be adjusted according to the selected resistor ratio.

The divider is configured as shown in the circuit diagram. The sensor can be powered using either a 5V external adaptor or the 5V/Vin pin when powered through USB. The ground is shared, and the midpoint of the divider connects

to an ADC-capable I/O pin on the board (pin 4 in this configuration).

EFY note. If the ZMPT101B voltage sensor has not been calibrated, refer to the calibration section to ensure accurate voltage readings.

Software

Within the firmware, the sensor pins are configured and the current sensor type (5A, 10A, or 30A) is selected. The voltage sensor pin and calibration parameters are also defined. Protection features are included to set over-voltage, under-voltage, and over-current thresholds, safeguarding the device and connected loads.

A provisioning mechanism connects the device to a Wi-Fi network for the end user. A dedicated system button enables manual data updates and synchronisation.